

Lecture 16: Optimal Consumption Problem

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Optimal Consumption Problem

Imagine you're a recent college graduate starting your career.
You have:

- A starting savings of \$10,000.
- An annual salary of \$60,000 that grows over time
- 40 years until retirement

Each year, you need to decide:

- How much to consume from your wealth.
- How to invest your remaining money across different assets.

Optimal Consumption Problem

You can choose between:

- A safe asset (like government bonds) with lower but predictable returns.
- Risky assets (like stocks) with higher average returns but more volatility.

Your goal is to maximize your lifetime happiness, which comes from:

- Enjoying consumption throughout your life.
- Leaving a bequest to your heirs.

Potential Mathematical Model

- plan consumption and investment at times $t = 0, \dots, T$
- in period t , wealth is k_t , consumption is c_t , labor income is y_t (all in inflation adjusted USD)
- remaining wealth invested in n assets with return mean μ and covariance Σ
- $r_t \in \mathbf{R}^n$ is asset return in period t
- $h_t \in \mathbf{R}^n$ denotes amounts invested in period t in USD
- $r_t^T h_t$ is the portfolio return in USD
- wealth dynamics given by

$$k_{t+1} = k_t - c_t + y_t + r_t^T h_t$$

Potential Mathematical Model

Maximize expected utility of consumption and bequest

$$\mathbf{E} \left(\frac{\beta}{\rho} k_T^\rho + \frac{1}{\rho} \sum_{t=0}^{T-t} c_t^\rho \right)$$

$\beta > 0$ sets relative importance of bequest; $\rho < 1$ sets risk aversion.

- a stochastic control problem, solved in [Merton, 1975]

Potential Mathematical Model

- replace stochastic wealth dynamics
 $k_{t+1} = k_t - c_t + y_t + r_t^T h_t$ with deterministic dynamics

$$k_{t+1} = k_t - c_t + y_t + \mu^T h_t - \frac{(1 - \rho)}{2} \frac{h_t^T \Sigma h_t}{k_t + v_t}$$

- v_t is the present value of future labor income discounted at the risk-free rate μ^{rf}

$$v_t = \sum_{\tau=t}^{T-1} y_{\tau} \exp(-\mu^{\text{rf}}(\tau - t))$$

- we require $k_t + v_t > 0$, i.e., wealth plus future labor income is positive
- last term is a pessimistic adjustment for risk derived in [Moehle and Boyd, 2021]

Potential Mathematical Model

- yields deterministic convex optimization problem

$$\begin{array}{ll}\text{maximize} & \frac{\beta}{\rho} k_T^\rho + \frac{1}{\rho} \sum_{t=0}^{T-1} c_t^\rho \\ \text{subject to} & k_{t+1} \leq k_t - c_t + y_t + \mu^T h_t - \frac{(1-\rho)}{2} \frac{h_t^T \Sigma h_t}{k_t + v_t} \\ & k_t = \mathbf{1}^T h_t, \quad c_t \geq 0\end{array}$$

(dynamic equality is replaced by inequality constraint, which is tight at solution)

- this certainty equivalent problem also solves stochastic problem
- can be extended to include mortality, liabilities, taxes, portfolio constraints, ...

Implementation

```
# Implement the Merton model with better scaling
k = cp.Variable(T+1) # Wealth at each time period (scaled)
h = cp.Variable((n, T)) # Amount invested in each asset at each time (scaled)
c = cp.Variable(T, nonneg=True) # Consumption at each time (scaled, non-negative)

# Cholesky decomposition of covariance matrix
try:
    Sigma_half = np.linalg.cholesky(Sigma)
except:
    # If Cholesky fails, use eigenvalue decomposition
    eigvals, eigvecs = np.linalg.eig(Sigma)
    Sigma_half = eigvecs @ np.diag(np.sqrt(np.maximum(eigvals, 0))) @ eigvecs.T

# Objective function (scaled)
objective = beta/rho * cp.power(k[T], rho) + 1/rho * cp.sum(cp.power(c, rho))

# Constraints
constraints = [k[0] == k0_scaled] # Initial wealth
constraints += [k[:-1] == cp.sum(h, axis=0)] # Wealth fully invested

# Add dynamics constraints
for t in range(T):
    risk_adjustment = (1-rho)/2 * cp.quad_over_lin(Sigma_half.T @ h[:,t], k[t] + v_scaled[t])
    constraints += [
        k[t+1] <= k[t] - c[t] + y_scaled[t] + mu.T @ h[:,t] - risk_adjustment,
        c[t] >= 0.001, # Minimum consumption to avoid numerical issues
        k[t] + v_scaled[t] >= 0.001 # Ensure positive denominator
    ]

# Form and solve problem
prob = cp.Problem(cp.Maximize(objective), constraints)
```


Implementation

```
# Try different solvers
try:
    prob.solve(solver=cp.ECOS, verbose=True)
except:
    try:
        prob.solve(solver=cp.SCS, verbose=True)
    except:
        try:
            prob.solve(solver=cp.CVXOPT, verbose=True)
        except:
            print("All solvers failed. Please check the problem formulation.")

if prob.status in ["infeasible", "unbounded"]:
    print(f"Problem is {prob.status}. Please check the constraints.")
else:
    print(f"Problem status: {prob.status}")
    print(f"Optimal value: ${prob.value * scale_factor:.2f}")

# Rescale back to original units
c_opt = c.value * scale_factor
k_opt = k.value * scale_factor
h_opt = h.value * scale_factor
```

Implementation

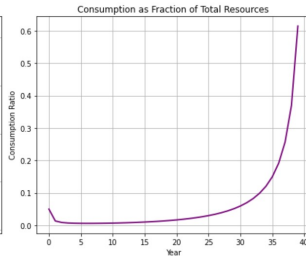
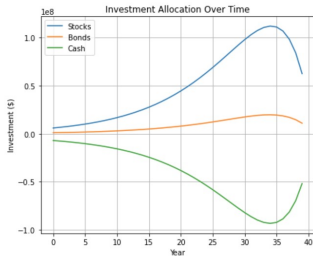
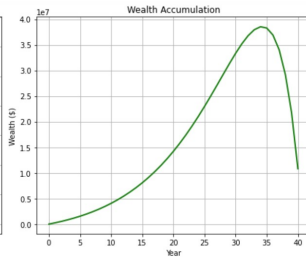
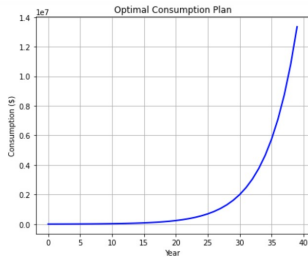
```
# Plot results
plt.figure(figsize=(12, 10))

# Consumption plan
plt.subplot(2, 2, 1)
plt.plot(range(T), c_opt, 'b-', linewidth=2)
plt.xlabel('Year')
plt.ylabel('Consumption ($)')
plt.title('Optimal Consumption Plan')
plt.grid(True)

# Wealth accumulation
plt.subplot(2, 2, 2)
plt.plot(range(T+1), k_opt, 'g-', linewidth=2)
plt.xlabel('Year')
plt.ylabel('Wealth ($)')
plt.title('Wealth Accumulation')
plt.grid(True)

# Investment allocation over time
plt.subplot(2, 2, 3)
asset_names = ['Stocks', 'Bonds', 'Cash']
for i in range(n):
    plt.plot(range(T), h_opt[i,:], label=asset_names[i])
plt.xlabel('Year')
plt.ylabel('Investment ($)')
plt.title('Investment Allocation Over Time')
plt.legend()
plt.grid(True)
```

Implementation



Summary Statistics:

Final wealth: \$10,811,969.08

Total consumption: \$70,147,682.33

Average consumption: \$1,753,692.06

Initial consumption: \$3,506.96

Final consumption: \$13,351,687.12

Young consume more?

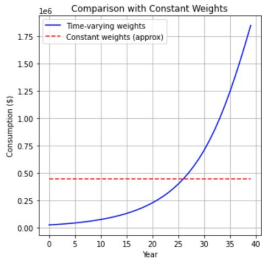
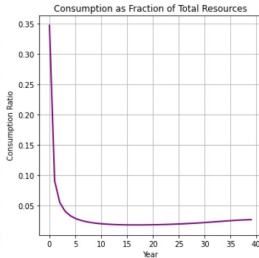
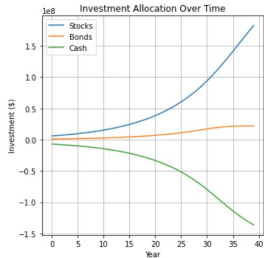
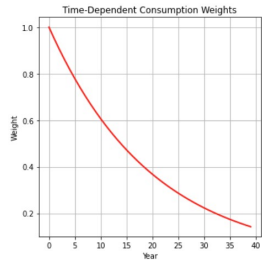
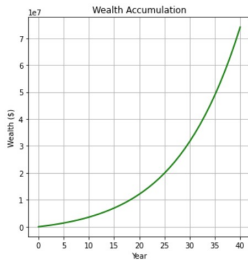
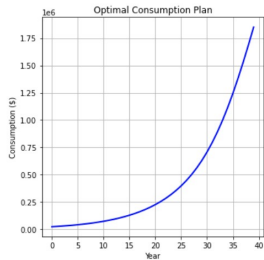
Time-Dependent Discount Factor

We can modify the objective function to have time-dependent weights:

$$\text{maximize } \frac{\beta}{\rho} k_T^\rho + \frac{1}{\rho} \sum_{t=0}^{T-1} \gamma_t c_t^\rho$$

where γ_t is a time-dependent weight that is higher in earlier periods.

Implementation



Summary Statistics:

Final wealth: \$74,077,945.41

Total consumption: \$17,899,473.69

Average consumption: \$447,486.84

Initial consumption: \$24,310.06

Final consumption: \$1,849,838.78

Consumption ratio (early/late): 0.01